



Conditional rehabilitation of cooperation under strategic uncertainty

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Abstract

In order to address the topic of the tragedy of the commons, more specifically that of tropical forest ecosystems explored as common-pool resources endowed with public-good features and exposed to deforestation risks, we consider game-theoretic population dynamics contingent on various differential equations. We propose an evolutionary model handed down to the Price theorem of selection. In a set of model-players evolving in strategic uncertainty and subject to certain mutation toward cooperation, the Price equation evens out unstructured and structured population replicator dynamics. According to the model outputs, avoiding the tragedy of the commons can be achieved on condition that half of the population temporarily exhibits a cooperative behavior. Furthermore, cooperative model-players ought to be rewarded at a level equivalent to the joint selection of cooperators and competitors issued from the unifying Price identity.

Keywords Bioeconomics · Evolutionary game theory · Price equation · Strategic uncertainty · Tragedy of the commons · Tropical forest management

Mathematics Subject Classification 34Dxx · 37Fxx · 37Hxx · 37Nxx · 91Axx · 91Cxx · 92Bxx

1 Introduction

The Guiana Shield, at the northern boundary of Amazonia, is one of the world-wide blocks of primary tropical forests (ONF 2014). The transboundary area embraces parts of northern Brazil, southern Venezuela, eastern Colombia and all of the Guianas

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(Guyana, Suriname and French Guiana). The size of continuous forest cover of 1,342,000 km² (Hammond 2005) corresponds to 26% of Amazonia (RAISG 2012). Through carbon storage and carbon sequestration, it holds a major role in mitigating climate change. Likewise, the rainforest provides ecosystemic services among which can be cited water regulation. It is also considered to contain significant levels of biodiversity (Bovolo et al. 2018). Due to the fact that countries forming the Guiana Shield are located within the same ecological region, they are considered to be responsible, before the international community, for the protection of the South American forest ecosystems (ONF 2014). According to the French National Forestry Office, the forests of the ecoregion are relatively stable and suffer from lower deforestation rates than other tropical forested areas. Nonetheless, strong economic and demographic levels of growth imply rising pressures on its natural ecosystems (ONF 2014). Although the public authorities have set themselves the objective to pursue their economic development in a sustainable manner, deforesting the ecoregion has the potential to cause multiple consequences at the local and regional scales (Bovolo et al. 2018). For example, forests are under threats of unsustainable logging, mining and of infrastructural development (Haden 1999). Furthermore, a wide-scale harvesting is carried through in the southern part of the Amazon basin, which reinforces the importance of the Guiana Shield as a valuable closed forest cover (Hammond 2005).

By avoiding deforestation and forest degradation, a cross-border collaboration to enhance climate mitigation of the Guiana Shield has been initiated in January 2013 through a REDD+ program. The project REDD+ for the Guiana Shield is the result of discussions of the regional states which have decided to carry out actions to those ends. Financing agreements have been provided by the European and French grant-financing funds (ONF 2014) as well as international organizations (GSF 2018). This type of action plan is intended to fight both against detrimental land use change and non-sustainable forest management. Equally, its purpose is to diminish greenhouse gas emissions, or to increase the forest carbon stock, coming from deforestation and forest degradation (UNFCCC 2019). Through rewarding efforts such as carbon credits or offsets generated by the avoidance of carbon emissions,¹ the set of countries in question agreed to promote low-carbon development. Cooperation is considered to be a key and a strategic component of the project, for it is meant to encourage partnership between neighboring countries. Yet, despite belonging to the same ecoregion, the member states have evolved in different contexts. Just as much, they have progressively set their own priorities and agendas (ONF 2014). In that manner, the displacement of gold mining or illegal logging activities, provoking leakage effects of deforestation between the member countries, have proved to be growing (ONF International 2015; Rahm et al. 2017).

By reason of the economic identification of forest areas, the international collective actions on forests continue to be challenging (Combes et al. 2016). Indeed, forests can be viewed as global public goods (Kaul et al. 2003), in that they are non-rival and non-excludable when it comes to mitigating carbon emissions (Hasson et al. 2010) or to preserving biodiversity (Deke 2008). Albeit, the main limitation to this standpoint is that it has little foundation in international law on rights (Combes et al. 2016).

¹ A carbon credit amounts to an emission reduction of one metric ton of equivalent carbon dioxide (CO₂e).

The other perspective is to consider forests as common goods, which are thus non-excludable but rivalrous among the population of users. It is in this way that the theory of common-pool resources, after which potential beneficiaries can be excluded owing to the common resource overuse, outlines the need for collective actions to solve local social dilemmas (Tucker 2010). Whereas the cooperation between the agents ought to be backed by complying with moderate levels of harvesting of a shared limited resource, its excessive exploitation leads to the well-known tragedy of the commons (Hardin 1968; Ostrom et al. 1999). This theory, derived from community-level studies, could be applied to grand systems as a solution to wide-scale common dilemmas (Fleischman et al. 2014). Those would be characterized by a system of governance of the commons over a large number of stakeholders evolving in a large spatial coverage (Keohane and Ostrom 1995). In order to prevent the tragedy of the commons, game-theoretic tools might be used to investigate on the incentives for the cooperation to emerge (Combes et al. 2016).

Unlike the social dilemmas related to the provision of public goods (Bach et al. 2006; Hauert et al. 2006; Santos et al. 2008; Wang et al. 2009; Dragicevic 2017), in which some agents called defectors free-ride on the provisions coming from the contributing agents, the social dilemmas revolving around the tragedy of the commons derive from the lack of coordination of the agents exploiting a shared resource. The dilemma between individual overharvesting and collective preservation often provokes the resource overuse and its eventual disappearance in the long run. In any case, the dilemma arises from the fact that non-cooperative defectors imperil the achievement of a higher collective payoff (Perc 2006). Nevertheless, the emanation of cooperation can be set off through spatial patterns (Nowak and May 1992), moral behavior (Capraro and Perc 2018), indirect reciprocity (Nowak and Sigmund 1998), voluntary interactions (Hauert et al. 2002), stochastic processes (Yoshimura and Jansen 1996) or through adaptive learning (Traulsen et al. 2004). Moreover, when game-theoretic strategies and the environment are both time-evolving (Perc and Szolnoki 2010), coevolutionary rules can help maintaining cooperative behaviors. In point of fact, evolutionary dynamics (Hofbauer and Sigmund 1998) is convenient for the study of global issues as it takes account of the whole population of agents. The coordinated outcome is then formalized with the help of law of mass-action (De Roos and Persson 2005). When envisioning the study of large populations undergoing stochastic phase transitions, statistical physics has also proved to be of great utility (Perc 2016; Capraro and Perc 2018).

In parallel, literature has addressed the importance of population structure. Models in population ecology often consist of unstructured and structured types (Odenbaugh 2005). Identical agents compose the population in the former case (De Roos and Persson 2005). In the latter case, agents are grouped into homogeneous classes. Structured populations are believed to better capture the reality of the situations than models built upon well-mixed sampling of players (Perc et al. 2017). Nowak and May (1992) showed that an evolutionary game in a structured population allows cooperators to avoid being exploited by defectors. Still, the emergence of cooperation and its evolutionary outcomes can depend on a number of other factors, among which can be quoted the structure of the interaction network and the type of interactions at stake (Perc et al. 2017). For instance, conditional cooperation depends on the number of cooperators already present in a group (Szolnoki and Perc 2012; Perc et al. 2013). If we now con-

sider an evolutionary game-theoretic setting where players ignore whether the game plan is cooperative or non-cooperative, for random payoff variations better represent longer-term contexts (Perc 2006), population dynamics should be considered under strategic uncertainty (Dragicevic 2019a).

In the present work, we borrow the tools from research on evolutionary game theory (Maynard and Price 1973). Provided that various evolutionary dynamics belong to a single unified framework (Page and Nowak 2002; Dragicevic 2019a), we have recourse to the Price version of replicator(–mutator) dynamics in a setting comprised of model-players sampled from both well- and poorly-mixed population settings. The Price theorem (Price 1970; Frank 1995; Kerr and Godfrey-Smith 2008) has been originally designed for the principle of selection of character states. Its purpose was to understand how the frequency of a particular character state changes over time (Dragicevic 2016, 2019b). The underlying model states that selection solely happens when character states covary with fitness or with the outcome they produce (Page and Nowak 2002; Knudsen 2004; Fox 2006; Collins and Gardner 2009; Helanterä and Uller 2010). The initial form of the equation is $\dot{E}(p) = \text{cov}(\pi, p) + E(\dot{p})$, where p incarnates the value of an arbitrary character state, $\dot{p}$ its time-evolution, and π the subsequent outcome. Indeed, the equation depends on $\text{cov}(\cdot)$ which describes selection.² As for $E(\cdot)$, it describes the environmental pressure argument on the evolution of the character-state value. Against the backdrop of our study, p is defined to be the probability of playing against a cooperator, whereas π stands for the payoff from playing a certain strategy. Combining the two enables us to study the expected payoff issued from selecting a particular strategy at the players' disposal. We thus translate selection $\text{cov}(\pi, p)$ into the distance between the expected payoff of a model-player and the average payoff in the population. Meanwhile, $E(\dot{p})$ underlines the payoff evolution with respect to the probability update. The expanded Price equation (Page and Nowak 2002; Page 2003), equally connected to the common evolutionary framework, yields $\dot{E}(p) = \text{cov}(\pi, p) + E(\dot{p}) + E(\pi \Delta)$. The meaning of the additional term is that of random mutation from a certain type of model-player to another.

The issue related to the public goods provision, built from the conflicts between contributing and defecting individuals, has already been raised through the Price equation (Zhang and Perc 2016). In their modeling framework, the last authors assume that the size of a group grows in size proportionally to its relative payoff. Selective migration of individuals toward a group takes place whenever its average payoff is higher. According to their results, the cooperation emerges when the inter-group selection, be it the first term of the equation, is greater than the intra-group selection defined by the second term. In order to address the topic of the tragedy of the commons, more specifically that of tropical forest ecosystems explored as common-pool resources endowed with public-good features and exposed to deforestation risks, we consider game-theoretic population dynamics contingent on various differential equations. We propose an evolutionary model handed down to the Price theorem of selection. In a set of model-players evolving in strategic uncertainty and subject to certain mutation toward cooperation, the Price equation evens out unstructured and structured popu-

² In the academic literature, selection is said to be one of the key mechanisms of evolution in which the differential survival, as a consequence of subsistence in a changing environment, depends on distinctions in individual character-state values.

lation replicator dynamics. According to the model outputs, avoiding the tragedy of the commons can be achieved on condition that half of the population temporarily exhibits a cooperative behavior. Furthermore, cooperative model-players ought to be rewarded at a level equivalent to the joint selection of cooperators and competitors issued from the unifying Price identity.

After this starting section, the static game plan under strategic uncertainty is introduced in Sect. 2. Section 3 provides a description of population dynamics examined by means of the Price identity. Section 4 is devoted to illustrating simulation examples. Section 5 concludes.

2 Static model

The standard game-theoretic model is as follows. Two model-players interact simultaneously. They can choose to play the cooperation strategy just as they can choose to be in competition, so that their types are not decided upon in advance. Let $w > 0.00$ be the model-player endowment or amount of wealth. Its contribution to cooperation, in order to secure the safeguard of the common good by inflicting a personal sacrifice, is given by c , with $c \leq w$. If model-players decide to cooperate, they both receive a reward that amounts to $\delta w > 0.00$, with $\delta \in [0.00, 1.00]$.

Model-players evolve in strategic uncertainty. This phenomenon lies within the lack of knowledge about their respective strategies. Subsequently, there is an unknown probability p of playing against a cooperator and an unknown probability $1.00 - p$ of playing against a competitor. Should both model-players decide to adopt the strategy that enables to preserve the common resource, the cooperative strategy salvages their wealth, which is reflected by $p(w(1.00 + \delta) - c)$. If they happen to be in competition and individually overuse the resource, they both receive the punishment payoff of $(1.00 - p)w$. When one of them plays the competition strategy while the other one has a cooperative behavior, the first receives the temptation payoff of pw ; the second is left with the so-called sucker's payoff of $(1.00 - p)(w - c)$.

2.1 Two-player game

The game payoff matrix originating from the prisoner's dilemma is shown in the table below.

	Cooperation	Competition
Cooperation	$w(1.00 + \delta) - c; w(1.00 + \delta) - c$	$w - c; w$
Competition	$w; w - c$	$w; w$

The payoffs of a model-cooperator and of a model-competitor within the context of strategic uncertainty are

$$\Pi(p) := \begin{cases} \pi_a = p(w(1.00 + \delta) - c) + (1.00 - p)(w - c) \\ \pi_b = pw + (1.00 - p)w. \end{cases} \quad (1)$$

When $p \in [0.00, 1.00]$, we have $\pi_a \gtrless \pi_b \Leftrightarrow p\delta w \gtrless c$. If $p\delta w > c$, that is, if the expected reward, obtained through the joint cooperative behavior, is greater than the costs of engaging in cooperation, cooperative strategies shall dominate the competitive ones in the game setting.

Proposition 1 *When the probability of playing against a cooperator is high enough, the mixed-strategy Nash equilibrium, in the context of strategic uncertainty, is to cooperate.*

Proof Straightforward from the payoff functions. $\square$

2.2 N-player game

Let us first extend the standard-game setting to N players, for the extension to a population-wide analysis is frequently used to study the issues on public goods (Santos et al. 2008; Perc et al. 2013). In that respect, we consider an infinitely large well-mixed population, also known as an unstructured type of population, which consists of fractions of cooperators x_a . The fraction of competitors is given by $x_b = 1.00 - x_a$. The fractions are meant to designate the sampled probabilities of choosing a particular strategy. We assume that $x_a + (1.00 - x_a) = 1.00$. The sum denotes a normalized population density in which 0.00 corresponds to extinction and 1.00 to full density. The overall population of players is supposed to be of size N . Now consider $\Sigma_i x_a$ and $\Sigma_j x_b$, with $i, j = 1, \dots, N$, to respectively identify the sizes of the population implicated in either playing the cooperative strategy or the competition strategy. On top of that, $\Sigma_i p = \Sigma_j p = 1.00$ represents the normalized probability of occurrence of strategies at the disposition of model-players. It follows that, for $\bar{p} = \Sigma_i x_a p$ and $i = 1, \dots, N$, $\Sigma_i x_a p \pi_a = \bar{p} \pi_a$ betokens the average expected population-wide payoff exhibiting a cooperative behavior. Model-players are sampled from the binomial distribution, such that the probability that there are n cooperators among $N - 1$ remaining players is of

$$\binom{N-1}{n} x_a^n (1.00 - x_a)^{N-1-n}. \quad (2)$$

Players compare their payoffs with that of a randomly chosen model-player from the population. The adoption of the strategy used by the model-player depends on a probability proportional to the difference between its payoff and their own. Given that $\sum_{n=0}^{N-1} \binom{N-1}{n} x_a^n (1.00 - x_a)^{N-1-n} = 1.00$, the expected payoffs of a model-cooperator

and of a model-competitor become

$$\Pi(p, x_a) := \begin{cases} \pi_a = p(w(1.00 + \delta) - c)x_a^{N-1} + (1.00 - p)(w - c) \\ \pi_b = pw + (1.00 - p)w(1.00 - x_a^{N-1}). \end{cases} \quad (3)$$

Through the introduction of x_a^{N-1} and of $1.00 - x_a^{N-1}$ into the payoff functions, we put the model-players in the context of strategic uncertainty, for either game plan can be at stake.

When $p \in [0.00, 1.00]$, we have $\pi_a \gtrless \pi_b \Leftrightarrow x_a \gtrless [(c + p(w - c))/(w + p(\delta w - c))]^{\frac{1}{N-1}}$. We thus have to compare the proportion of cooperators in a well-mixed population with the cost-benefit ratio increased by the expected net benefit from cooperating.

Should the probability of playing against a cooperator be unknown, the cooperative strategy shall dominate in mixed strategies would the fraction of cooperators tend to full density. In the opposite case, competition will dominate in the game setting. Specifically, when $c \rightarrow w$, competition will always prevail. When the cost of engaging in cooperation tends to zero ($c \rightarrow 0.00$), the preponderance of cooperators in the population suffices to propagate the strategy of cooperation among the competitors. In brief, albeit the probability of playing against a certain type of model-player decides on the game resolution, its influence significantly diminishes in the face of the population fractions already playing a strategy.

Proposition 2 *When the preponderance of cooperators in a well-mixed population is vast enough, the mixed-strategy Nash equilibrium, in an N -player game under strategic uncertainty, is to cooperate.*

Proof Straightforward from the payoff functions. $\square$

3 Price identity dynamic model

3.1 Price identity of unstructured population replicator dynamics

Now consider the N -player game composed of model-players, called replicator, that co-evolve in existing types. Following the replicator dynamics (Hofbauer and Sigmund 1998), their evolution is given by the following system of differential equations

$$\dot{X} := \begin{cases} \dot{x}_a = x_a(\pi_a - \bar{\pi}) \\ \dot{x}_b = (1.00 - x_a)(\pi_b - \bar{\pi}). \end{cases} \quad (4)$$

Within the system, and at any given point of time, x_a and $1.00 - x_a$ stand for the densities of payoffs of a model-cooperator and of a model-competitor. The frequency of changes of their strategies depends on a per capita rate. The last is equal to the difference between the expected payoff of a model-player and the average payoff recorded in the population (Cressman and Tao 2014). Those payoffs result from the

interactions taking place in groups of individuals formed purposely in a random way. The expression $\bar{\pi} = x_a\pi_a + x_b\pi_b = x_a\pi_a + (1.00 - x_a)\pi_b$ indicates the average payoff in the population.

The system can be rewritten as a Price equation in the following form

$$\begin{aligned}
 \dot{E}(p) &= cov(\pi_a, p) + E(\dot{p}) \\
 &= \sum_i x_a \pi_a p - \bar{\pi} \bar{p} + \sum_i x_a \dot{p} \\
 &= \sum_i x_a \pi_a p - (x_a \pi_a + (1.00 - x_a) \pi_b) \sum_i p x_a + \sum_i x_a \dot{p} \\
 &= \sum_i p x_a ((1.00 - x_a) \pi_a - (1.00 - x_a) \pi_b) + \sum_i x_a \dot{p} \\
 &= \sum_i p x_a ((1.00 - x_a) (\pi_a - \pi_b)) + \sum_i x_a \dot{p} \\
 &= x_a (1.00 - x_a) (\pi_a - \pi_b) \sum_i p x_a + \sum_i x_a \dot{p} \\
 &= x_a (1.00 - x_a) \left[p (w(1.00 + \delta) - c) x_a^{N-1} + (1.00 - p) (w - c) \right] \sum_i p x_a \\
 &\quad + x_a (1.00 - x_a) \left[-pw - (1.00 - p)w (1.00 - x_a^{N-1}) \right] \sum_i p x_a + \sum_i x_a \dot{p},
 \end{aligned} \tag{5}$$

where $n \in i = 1, \dots, N$ and $\dot{p} = n \bar{p} (N + 5.00 \times 10^{-3} \frac{n}{N}) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$.³ The properties of the differential equation lead to the following proposition.

Proposition 3 *In unstructured population replicator dynamics under strategic uncertainty – handed down to the Price theorem of selection –, competition as an equilibrium undergoes cyclical behavior. In presence of risk-neutral model-players, full density of cooperators could nonetheless be attained. In that case, the former ought to occur transitorily, by dint of cyclical shifts, at the expense of their diminished levels of wealth.*

Proof The proof can be found in the appendices section. $\square$

3.2 Price identity of structured population replicator dynamics

By introducing an update decay in the comparison of payoffs with those of randomly chosen model-players (Sato and Crutchfield 2003), we shall now shift from an unstructured configuration of the population to a structured one. The average payoff is now weighted by the relative proportion of players that inhere in the adverse strategy. This means that the binomial sampling becomes performed from a poorly-mixed population of model-players. It corresponds to the aftereffect of replication unfolded in disproportionate population fractions. It then could be that the spread of a particular strategy among model-players is lowered. The respective decay parameters of cooperators and competitors are expressed via $\alpha = (1.00 - x_a)/x_a$ and $\beta = x_a/(1.00 - x_a)$. As a corollary, replicator dynamics correspond to a case of update decay where $x_a = 1.00 - x_a$. That situation brings us back to a well-mixed population sampling.

³ The expression 5.00×10^{-3} represents the increment rate for the bifurcation analysis that will follow.

The system evolution across the time is equivalent to

$$\dot{X}(\alpha, \beta) := \begin{cases} \dot{x}_a(\alpha) = x_a (\pi_a - \alpha \bar{\pi}) \\ 1 - \dot{x}_a(\beta) = (1.00 - x_a) (\pi_b - \beta \bar{\pi}) \end{cases} \quad (6)$$

The differential equations yield a single Price formulation in form of

$$\begin{aligned} \dot{E}(\alpha, p) &= cov(\pi_a, p) + E(\dot{p}) \\ &= \Sigma_i x_a \pi_a p - \alpha \bar{\pi} \bar{p} + \Sigma_i x_a \dot{p} \\ &= \Sigma_i x_a \pi_a p - \frac{1.00 - x_a}{x_a} (x_a \pi_a + (1.00 - x_a) \pi_b) \Sigma_i p x_a + \Sigma_i x_a \dot{p} \\ &= \Sigma_i p x_a \left(\pi_a x_a - \frac{(1.00 - x_a)^2}{x_a} \pi_b \right) + \Sigma_i x_a \dot{p} \\ &= \Sigma_i p x_a \left(\frac{x_a^2 \pi_a - (1.00 - x_a)^2 \pi_b}{x_a} \right) + \Sigma_i x_a \dot{p} \\ &= \Sigma_i p \left(x_a^2 \pi_a - (1.00 - x_a)^2 \pi_b \right) + \Sigma_i x_a \dot{p} \\ &= \left[x_a^2 \left(p (w(1.00 + \delta) - c) x_a^{N-1} + (1.00 - p)(w - c) \right) \right] \Sigma_i p \\ &\quad + \left[-(1.00 - x_a)^2 \left(p w + (1.00 - p) w \left(1.00 - x_a^{N-1} \right) \right) \right] \Sigma_i p + \Sigma_i x_a \dot{p}, \end{aligned} \quad (7)$$

where $n \in i = 1, \dots, N$ and $\dot{p} = n \bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$.

The following proposition ensues.

Proposition 4 *In structured population replicator dynamics under strategic uncertainty—handed down to the Price theorem of selection, competition as an equilibrium can be stable, unstable or undergo cyclical behavior. In presence of risk-neutral model-players, full density of cooperators could nonetheless be attained. In that case, the former ought to occur transitorily, by dint of cyclical shifts, at the expense of their diminished levels of wealth, which can end up being null.*

Proof The proof can be found in the appendices section. $\square$

3.3 Price identity of unstructured population replicator–mutator dynamics

The probability of a change in strategy between two points in time echoes the happening of mutation (Chatterjee et al. 2012), the rate of which is given by $q_{ji} \in [0.00, 1.00]$. This brings to the fore a differential equation, which describes the adaptive dynamics of the population distribution (Levin 2002), termed replicator–mutator dynamics with a frequency-dependent fitness (Bürger 1998). Mutation mostly arises when groups of players form alliances in order to increase their payoffs (Nolte 2015). When it comes to the topic addressed in this paper, a competitive strategist then replicates into a cooperative one (Dercole and Rinaldi 2008). The mutation is supposed to give an

incentive to a model-player to avoid its individual overharvesting of a joint natural resource. Namely, the player is encouraged to moderate its exploitation of a common good supplying public good benefits to the overall population.

The population evolution, as defined in the Price equation with mutational events (Page and Nowak 2002), takes the form of

$$\begin{aligned}
 \dot{E}(\Delta, p) &= \text{cov}(\pi_b, p) + E(\dot{p}) + E(\pi_a \Delta) \\
 &= \sum_j x_b \pi_b p - \bar{\pi} \bar{p} + \sum_i x_a \dot{p} + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= \sum_j x_b \pi_b p - (x_a \pi_a + (1.00 - x_a) \pi_b) \sum_i p x_a + \sum_i x_a \dot{p} + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= \sum_j p \pi_b x_b - \sum_i x_a^2 \pi_a p - \sum_i x_a (1.00 - x_a) \pi_b p + \sum_i x_a \dot{p} + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= -\sum_i x_a^2 \pi_a p + \sum_i x_a \dot{p} + \pi_b p (\sum_j x_b - \sum_i x_a (1.00 - x_a)) + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= \sum_i x_a \pi_a p (\sum_j q_{ji} - x_a) + \pi_b p (\sum_j x_b - \sum_i x_a (1.00 - x_a)) + \sum_i x_a \dot{p} \\
 &= \sum_i x_i \left(p (w(1.00 + \delta) - c) x_a^{N-1} + (1.00 - p) (w - c) \right) p (\Delta - x_a) \\
 &\quad + \left(p w - (1.00 - p) w (1.00 - x_a^{N-1}) \right) p (\sum_j x_b + \sum_i x_a (1.00 - x_a)) \\
 &\quad + \sum_i x_a \dot{p}, \tag{8}
 \end{aligned}$$

where $i, j = 1, \dots, N$, $\Delta = 1.00^4$ and $\dot{p} = n \bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$ for $n \in i = 1, \dots, N$.

Proposition 5 *In unstructured population replicator–mutator dynamics under strategic uncertainty—handed down to the Price theorem of selection, competition turns out to be a steady state. In presence of risk-neutral model-players, full density of cooperators could nonetheless be attained. In that case, the former occurs would the likelihood of coming across a cooperator be high enough.*

Proof The proof can be found in the appendices section. $\square$

It has been proved that, in replicator–mutator dynamics, the strategy replication under certain mutation in an unstructured population is equivalent to a negative strategy replication in a structured population (Dragicevic 2019a). Despite the fact that the Price equation belongs to a single evolutionary framework, the results obtained in terms of reflective evolution are no longer observed. By combining the above-noted result with that obtained in the previous subsection, together with considering the certainty of mutation by posing $E(\Delta) = 1.00$,⁵ the following conditional equivalence follows up

$$\begin{aligned}
 &\left(x_a^2 \pi_a - (1.00 - x_a)^2 \pi_b \right) p \sum_i x_a + E(\dot{p}) \\
 &= (\text{cov}(\pi_b, p) + \bar{\pi} \bar{p})(\Delta - x_a) + \pi_b p (\sum_j x_b - \sum_i x_a (1.00 - x_a)) + E(\dot{p})
 \end{aligned}$$

⁴ As a result of prevalence proportion (Dragicevic 2017), we know that the mutating probability of a player—randomly picked from the population—is equal to the cooperative proportion of the population. When the population-scale mutation from type j to type i takes place, $E(\pi_a \Delta)$ denotes the expected surplus from cooperating. The certainty of mutation equals to having $E(\Delta) = 1.00$.

⁵ Having in mind that $E(\Delta) = 1.00$ is equivalent to having $E(\dot{p}) \rightarrow 0.00$, that is, the expected change of p tends to zero once the population of players moves toward cooperation on a certain basis.

$$\begin{aligned}
p \Sigma_i x_a \pi_a &= \frac{((\text{cov}(\pi_b, p) + \bar{\pi} \bar{p})(\Delta - x_a) + \pi_b p(\Sigma_j x_b - \Sigma_i x_a(1.00 - x_a)))\pi_a}{x_a^2 \pi_a - (1.00 - x_a)^2 \pi_b} \\
p \Sigma_i x_a \pi_a &= \frac{((\text{cov}(\pi_b, p) + \bar{\pi} \bar{p})(\Delta - x_a) + \pi_b p(\Sigma_j x_b - \Sigma_i x_a(1.00 - x_a)))\pi_a}{x_a^2 \pi_a - (1.00 - x_a)^2 \pi_b} \\
x_a^2 \pi_a - (1.00 - x_a)^2 \pi_b &= \frac{((\text{cov}(\pi_b, p) + \bar{\pi} \bar{p})(\Delta - x_a) + \pi_b p(\Sigma_j x_b - \Sigma_i x_a(1.00 - x_a)))\pi_a}{p \Sigma_i x_a \pi_a} \\
\dot{x}_a(\alpha) &= \frac{((\text{cov}(\pi_b, p) + \bar{\pi} \bar{p})(\Delta - x_a) + \pi_b p \Sigma_j x_b - \pi_b p \Sigma_i x_a(1.00 - x_a))\pi_a}{p \Sigma_i x_a \pi_a} \\
\dot{x}_a(\alpha) &= \frac{((\text{cov}(\pi_b, p) + \bar{\pi} \bar{p})(1.00 - x_a) + \pi_b p \Sigma_j x_b - \pi_b p \Sigma_i x_a(1.00 - x_a))\pi_a}{\text{cov}(\pi_a, p) + \bar{\pi} \bar{p}} \\
\dot{x}_a(\alpha) &= \frac{((\text{cov}(\pi_a, p) + \bar{\pi} \bar{p})(1.00 - x_a) + (\text{cov}(\pi_b, p) + \bar{\pi} \bar{p})(1.00 - x_a))\pi_a}{\text{cov}(\pi_a, p) + \bar{\pi} \bar{p}} \\
\dot{x}_a(\alpha) &= \frac{\pi_a(1.00 - x_a)(\text{cov}(\pi_a, p) + \bar{\pi} \bar{p} + \text{cov}(\pi_b, p) + \bar{\pi} \bar{p})}{\text{cov}(\pi_a, p) + \bar{\pi} \bar{p}} \\
\dot{x}_a(\alpha) &= \frac{\pi_a(1.00 - x_a)(\text{cov}(\pi_a + \pi_b, p) + 2\bar{\pi} \bar{p})}{\text{cov}(\pi_a, p) + \bar{\pi} \bar{p}} \\
\dot{x}_a(\alpha) &= \frac{\pi_a(1.00 - x_a)}{\text{cov}(\pi_a, p) + \bar{\pi} \bar{p}} (\text{cov}(\pi_a + \pi_b, p) + 2\bar{\pi} \bar{p}). \tag{9}
\end{aligned}$$

The next proposition ensues.

Proposition 6 *In replicator–mutator dynamics under strategic uncertainty—handed down to the Price theorem of selection, the unstructured-population strategy replication under certain mutation from competition to cooperation is conditionally equivalent to the structured-population strategy replication exempt from mutation. The sameness comes into being would the fraction of competitors, weighted by the payoff of a cooperative model-player and inversely proportioned to the expected payoff from cooperation, be less than or equal to the joint selection of cooperators and competitors increased by the double average population payoff.*

Proof The proof can be found in the appendices section. $\square$

It must be highlighted that the subtraction of the environmental pressure from the argument devoted to selection suggests that the former is implicitly incorporated in the latter. Such a mathematical formalization goes beyond the original form and interpretation of the Price theorem, the expression of which could eventually have made redundant the inclusion of external factors. Moreover, this claim does not invalidate the outcome presented in Melbinger and Vergassola (2015). The last authors point out that the environmental variability influences selection through what they term neutral evolution. It underscores that the evolutive fitness, which exhibits the updating payoff of a model-player in our modeling setting, implicitly varies with respect to already incorporated environmental fluctuations.

The unexpected outcome issued from the unification of structured population replicator dynamics and of unstructured population replicator–mutator dynamics comes from the appearance of $\dot{x}_a(\alpha)$, that is, from the fact that standard replicator equation applied to a poorly-mixed population setting liaises between the differential equations at stake. As a matter of fact, the findings in Dragicevic (2019a) concede that standard replication only appears as a reflection of structured population replicator–mutator dynamics. In the present Price identity setting, the dynamics is found not to be reflective. Instead, it is subject to a modified version of selection, which gives, in respect to the population kind of configuration, an all-inclusive Price reading of evolutionary dynamics.

Remark 1 When it comes to the Price theorem of selection under strategic uncertainty, the equivalence between structured population replicator dynamics and unstructured population replicator–mutator dynamics can be established, in an indirect manner, by the calibration of replicator dynamics taking place in a poorly-mixed population setting.

3.4 Price identity of structured population replicator–mutator dynamics

In the lookout for the structured population version of the differential equation, the replicator–mutator dynamics takes place in the form of

$$\begin{aligned}
 \dot{E}(\alpha, \Delta, p) &= cov(\pi_b, p) + E(\dot{p}) + E(\pi_a \Delta) \\
 &= \sum_j x_b \pi_b p - \alpha \bar{\pi} \bar{p} + \sum_i x_a \dot{p} + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= \sum_j x_b \pi_b p - \frac{1.00 - x_a}{x_a} (x_a \pi_a + (1.00 - x_a) \pi_b) \sum_i p x_a \\
 &\quad + \sum_i x_a \dot{p} + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= \sum_j p x_b - (1 - x_a) \pi_a \sum_i p x_a - \frac{(1 - x_a)^2}{x_a} \pi_b \sum_i p x_a \\
 &\quad + \sum_i x_a \dot{p} + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= \sum_j p x_b - \sum_i x_a (1 - x_a) \pi_a \sum_i p x_a - \frac{(1 - x_a)^2}{x_a} \pi_b \sum_i p x_a \\
 &\quad + \sum_i x_a \dot{p} + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= -\sum_i x_a (1.00 - x_a) \pi_a p + \sum_i x_a \dot{p} + \pi_b p (\sum_j x_b - \sum_i (1.00 - x_a)^2 p) \\
 &\quad + \sum_i x_a \pi_a \sum_j q_{ji} p \\
 &= \sum_i x_a \pi_a p (\sum_j q_{ji} - (1.00 - x_a)) + \pi_b p (\sum_j x_b - \sum_i (1.00 - x_a)^2 p) \\
 &\quad + \sum_i x_a \dot{p} \\
 &= \sum_i x_a \left(p (w(1.00 + \delta) - c) x_a^{N-1} \right. \\
 &\quad \left. + (1.00 - p) (w - c) p (\Delta - (1.00 - x_a)) \right. \\
 &\quad \left. + \left(p w - (1.00 - p) w (1.00 - x_a^{N-1}) \right) p (\sum_j x_b - \sum_i (1.00 - x_a)^2 p) \right. \\
 &\quad \left. + \sum_i x_a \dot{p}, \right. \tag{10}
 \end{aligned}$$

where $i, j = 1, \dots, N$ and $\dot{p} = n\bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$ for $n \in i = 1, \dots, N$.

Proposition 7 *In structured population replicator–mutator dynamics under strategic uncertainty—handed down to the Price theorem of selection, competition is an equilibrium that can be stable as well as subject to cyclical shifts. In presence of risk-neutral model-players, full density of cooperators could nonetheless be attained. In that case, the former occurs would the probability of playing against a cooperator be of one or would the cost of cooperating be equal to the rewarding multiplier.*

Proof The proof can be found in the appendices section. $\square$

The previous result goes along with that of Capraro and Perc (2018) who explain that the higher the multiplication factor, the higher the tolerance toward the number of defectors in a group. Quite naturally, if the cost is equal to the rewarding multiplier, the rewarding policy shall never produce enough cooperation.

In the same vein, Dragicevic (2019a) has shown that in replicator–mutator dynamics with strategic uncertainty and update decay, the strategy replication under certain mutation in a structured population is equivalent to a negative replication issued from an unstructured population. Reflective evolution within the Price identity setting is no valid this time either. Along with the assumption that mutation occurs for sure, extending the analysis by merging the findings obtained in Sect. 3.1. and in this one establishes the subsequent conditional equivalence

$$\begin{aligned}
 & x_a(1.00 - x_a)(\pi_a - \pi_b)p\Sigma_i x_a + E(\dot{p}) \\
 &= (cov(\pi_b, p) + \bar{\pi}\bar{p})(\Delta - x_a) + \pi_b p (\Sigma_j x_b - \Sigma_i x_a(1.00 - x_a)) + E(\dot{p}) \\
 p\Sigma_i x_a \pi_a &= \frac{((cov(\pi_b, p) + \bar{\pi}\bar{p})(\Delta - x_a) + \pi_b p (\Sigma_j x_b - \Sigma_i x_a(1.00 - x_a)))\pi_a}{x_a(1.00 - x_a)(\pi_a - \pi_b)} \\
 p\Sigma_i x_a \pi_a &= \frac{((cov(\pi_b, p) + \bar{\pi}\bar{p})(\Delta - x_a) + \pi_b p (\Sigma_j x_b - \Sigma_i x_a(1.00 - x_a)))\pi_a}{x_a(1.00 - x_a)(\pi_a - \pi_b)} \\
 &= \frac{x_a(1.00 - x_a)(\pi_a - \pi_b)}{((cov(\pi_b, p) + \bar{\pi}\bar{p})(\Delta - x_a) + \pi_b p (\Sigma_j x_b - \Sigma_i x_a(1.00 - x_a)))\pi_a} \\
 &= \frac{p\Sigma_i x_a \pi_a}{((cov(\pi_b, p) + \bar{\pi}\bar{p})(\Delta - x_a) + \pi_b p (\Sigma_j x_b - \Sigma_i x_a(1.00 - x_a)))\pi_a} \\
 \dot{x}_a &= \frac{((cov(\pi_b, p) + \bar{\pi}\bar{p})(\Delta - x_a) + \pi_b p \Sigma_j x_b - \pi_b p \Sigma_i x_a(1.00 - x_a))\pi_a}{p\Sigma_i x_a \pi_a} \\
 \dot{x}_a &= \frac{((cov(\pi_b, p) + \bar{\pi}\bar{p})(1.00 - x_a) + \pi_b p \Sigma_j x_b - \pi_b p \Sigma_i x_a(1.00 - x_a))\pi_a}{cov(\pi_a, p) + \bar{\pi}\bar{p}} \\
 \dot{x}_a &= \frac{((cov(\pi_a, p) + \bar{\pi}\bar{p})(1.00 - x_a) + (cov(\pi_b, p) + \bar{\pi}\bar{p})(1.00 - x_a))\pi_a}{cov(\pi_a, p) + \bar{\pi}\bar{p}} \\
 \dot{x}_a &= \frac{\pi_a(1.00 - x_a)(cov(\pi_a, p) + \bar{\pi}\bar{p} + cov(\pi_b, p) + \bar{\pi}\bar{p})}{cov(\pi_a, p) + \bar{\pi}\bar{p}}
 \end{aligned}$$

$$\begin{aligned}\dot{x}_a &= \frac{\pi_a(1.00 - x_a)(\text{cov}(\pi_a + \pi_b, p) + 2\bar{\pi}\bar{p})}{\text{cov}(\pi_a, p) + \bar{\pi}\bar{p}} \\ \dot{x}_a &= \frac{\pi_a(1.00 - x_a)}{\text{cov}(\pi_a, p) + \bar{\pi}\bar{p}}(\text{cov}(\pi_a + \pi_b, p) + 2\bar{\pi}\bar{p}).\end{aligned}\quad (11)$$

We summarize the result in the following proposition.

Proposition 8 *In replicator–mutator dynamics under strategic uncertainty—handed down to the Price theorem of selection, the structured-population strategy replication under certain mutation from competition to cooperation is conditionally equivalent to the unstructured-population strategy replication exempt from mutation. The same-ness comes into being would the fraction of competitors, weighted by the payoff of a cooperative model-player and inversely proportioned to the expected payoff from cooperation, be less than or equal to the joint selection of cooperators and competitors increased by the double average population payoff.*

Proof The proof can be found in the appendices section. $\square$

The above-mentioned proposition partly meets the result pointed out by Frank (1995) and by Zhang and Perc (2016). It does emphasize, in the Price pattern of selection, on between-group pressure and on within-group pressure. Nevertheless, the between-group pressure both considers cooperators and competitors within each group. Likewise, the within-group pressure, portrayed by the expected payoff from cooperation, becomes inversely proportioned rather than additive. The novel feature from the foregoing results thus lies in the ability to bridge two separate models of population dynamics. As a matter of fact, the unintended outcome issued from the uniting of unstructured population replicator dynamics and of structured population replicator–mutator dynamics stems from the manifestation of $\dot{x}_a$. In other words, standard replicator equation applied to a well-mixed population setting is in capacity to bring together two polarized differential equations, be it a result that does not appear in the reflective dynamics found out in Dragicevic (2019a).

Remark 2 When it comes to the Price theorem of selection under strategic uncertainty, the equivalence between unstructured population replicator dynamics and structured population replicator–mutator dynamics can be established, in an indirect manner, by a calibration of replicator dynamics taking place in a well-mixed population setting.

The expressions (9) and (11) enable us to determine the fraction of cooperators by which the unification of evolutionary dynamics can be envisaged. We thus obtain

$$\begin{aligned}\dot{x}_a(\alpha) &= \dot{x}_a \\ x_a^2\pi_a - (1.00 - x_a)^2\pi_b &= x_a(1.00 - x_a)(\pi_a - \pi_b) \\ x_a^* &= 0.50.\end{aligned}\quad (12)$$

The ultimate analysis of the game-theoretic setting gives rise to the following corollary.

Corollary 1 *In a set of model-players evolving in strategic uncertainty and subject to certain mutation toward cooperation, the Price theorem of selection evens out unstructured population replicator dynamics and structured population replicator dynamics on condition that half of the population exhibits a cooperative behavior.*

4 Simulations

Starting from the propositions obtained in the modeling section, we shall now illustrate, by means of numerical simulations performed on bifurcation diagrams, the various evolutionary dynamics. Four examples are presented, each of which covers the properties and the conditions exposed hereinabove. The purpose of the incremental fifth subsection is to bring to light the level of the weighted aggregated selection of cooperators and competitors.

In what follows, consider the initial numerical values of the modeling items to be fixed at $w = 10.00$, $\delta = 0.20$, $c = 5.00$, $N = 99$, $p = 0.50$ and $q_{ji} = 1.00$. As we now launch the simulations, the dynamic mappings yield the following outputs.

4.1 Price identity of unstructured population replicator dynamics

The case of replicator dynamics, applied to an unstructured population setting, is illustrated in Fig. 1.

We shall now focus more specifically on the case where $p = 0.25$. The latter unveils three main patterns. The first pattern is at $\dot{E}(0.25) = 0.00$ and corresponds to the straight line connecting $x_a = 0.00$ to $x_a = 1.00$. In the absence of an interior equilibrium, the perfect order is in the form of cyclicity between the absence of cooperative model-players and the attainment of full cooperation in the population. The second pattern denotes the increasing system order having the aspect of s-shaped curves. Their origin stands at $x_a = 0.00$ where $\dot{E}(0.25) = 0.00$. The pattern reaches the extreme value of $\dot{E}(0.25) = -16.31$ at $x_a = 0.74$. It can be mentioned that the inflection point of the left-sided sigmoid portion of the bifurcation diagram has for coordinate $x_a = 0.50$, such that the trajectory conversion toward full cooperation happens to occur when the population is equally divided between competitors and cooperators. The third pattern is to be considered as that in which the system metastability goes through a series of periodic orbits. Those take the form of non-monotonic u-shaped curves which all bifurcate toward 0.00 or 1.00. At $x_a = 0.50$, a coordinate at which the time derivative among the trajectories is minimal, is the least unstable equilibrium. In brief, while multiple bifurcations display the fixed point instability at $x_a = 1.00$, the convergence to corner equilibria shows decreasing chaos in the direction of full competition.

The numerical values confirm the investigation of the second derivative from which the inflection points have been found to stand at $x_a = 0.00$ and at $x_a = 0.74$. In the second case, $F''(x_a) = -16.31$ yields $w = 6.33$, be it a value similar to that of $w = \frac{c}{1.00+\delta}$. This implies that the inflection point takes place when the initial endowment is brought forward to the discounted value of the expense of cooperation.

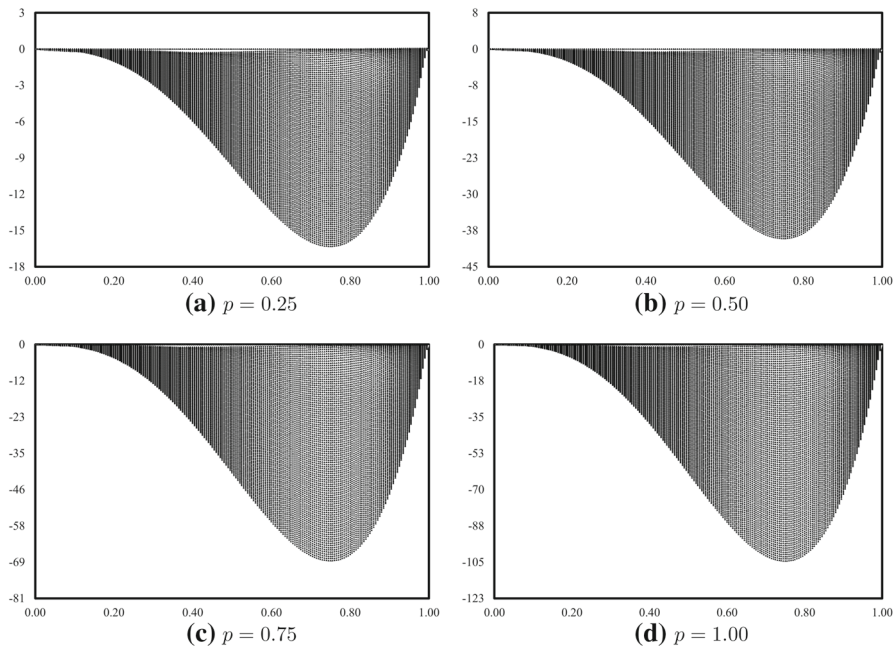


Fig. 1 Bifurcation diagrams of the Price identity applied to unstructured population replicator dynamics ($\dot{E}(p)$) with increment rate of 5.00×10^{-3} and four probability levels defining the likelihood of cooperation ($p = \{0.25, 0.50, 0.75, 1.00\}$). The x -axes correspond to the proportions of cooperators (x_a). The y -axes denote differential equation (5) as an implicit function of time ($E(p)/dt$). The model generates three main patterns of population dynamics. The trajectories to fixed points are illustrated through straight lines and periodic orbits, the latter being represented by a series of bell-shaped and of s-shaped curves. They exhibit increasing order close to full competition and cyclic temporary bifurcations toward full cooperation

The same interpretation of numerical results can be applied to the remaining probability levels. The noticeable difference is to be found in higher shifts in population fractions devoted to cooperation, for the time derivative reaches the peak value, when $p = 1.00$, of $\dot{E}(1.00) = -104$ at $x_a = 0.77$.

4.2 Price identity of structured population replicator dynamics

The scenario of structured population replicator dynamics (Fig. 2) unveils patterns reasonably different from those of standard replicator dynamics.

The emphasis is once again laid on the initial case of $p = 0.25$. The first pattern is at $\dot{E}(\alpha, 0.25) = 0.00$ and corresponds to the same cyclical shift, in perfect order, from extinction to full density of cooperators. The second pattern, revealing the system metastability, is in form of non-monotonic u-shaped curves which all bifurcate toward 0.00 or 1.00. The intersections of their trajectories with the straight line occur at $x_a = 0.00$ and at $x_a = 0.61$. Two separate regions, shaped by a series of paths, form the bifurcation areas in the overall diagram. The trend toward a population fully composed of competitors, illustrated by a pond-like region, occurs at a moderate pace, for the negativity of the time derivative reaches the extreme value, at $x_a = 0.30$, of

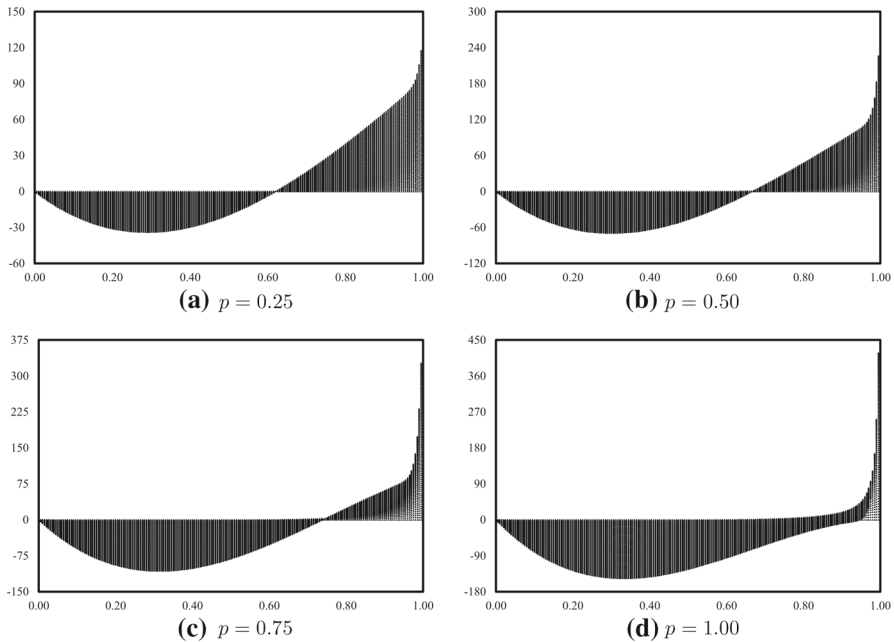


Fig. 2 Bifurcation diagrams of the Price identity applied to structured population replicator dynamics ($\dot{E}(\alpha, p)$) with increment rate of 5.00×10^{-3} and four probability levels defining the likelihood of cooperation ($p = \{0.25, 0.50, 0.75, 1.00\}$). The x -axes correspond to the proportions of cooperators (x_a). The y -axes denote differential equation (7) as an implicit function of time ($E(\alpha, p)/dt$). The model generates three main patterns of population dynamics. The trajectories to fixed points are illustrated through straight lines and periodic orbits, the latter being represented by a series of bell-shaped and of s-shaped curves. They exhibit increasing order close to full competition and cyclic temporary bifurcations toward full cooperation

$\dot{E}(0.25) = -34.00$. In this instance, the initial level of wealth equal to 10.00 reaches the value of 52.82, that is, the initial endowment is increased fivefold. In other words, in order to switch either to full competition or to full cooperation, the level of wealth that orients toward either population-level extremum is well above the initial allocation.

In contrast, the triangular area depicting convergence toward full density of cooperators is sharper, with an acute change of slope at $x_a = 0.93$. The maximum instability value to the cooperative population setting, recorded at $x_a = 1.00$, is of $\dot{E}(0.25) = 118$. So it can be seen that, in absolute terms, the instability toward full cooperation is much greater than the level of stability toward its extinction. As the likelihood toward cooperation increases, greater levels of x_a distinguish between two attraction areas. Another interesting feature is that, as $p \rightarrow 1.00$, the portion of the negative orthant containing the basin-shaped area increases both in size and stability. Instead, triangular areas representing the spaces in which cooperation takes place have their sized reduced. Nonetheless, their smaller basins of attraction become gradually located in both negative and positive orthants. In consequence, heading toward the certainty of cooperation translates into obtaining greater stability of population-wide cooperation. Yet, the attainment of $x_a = 1.00$ can only be achieved asymptotically, for the time derivative reaches the value of $\dot{E}(1.00) = 418$.

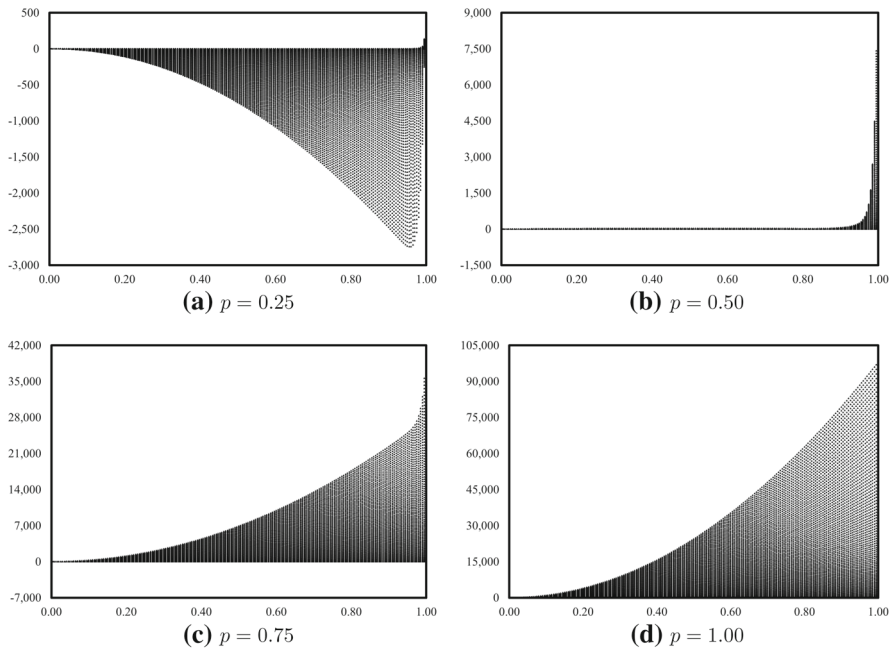


Fig. 3 Bifurcation diagrams of the Price identity applied to unstructured population replicator–mutator dynamics ($\dot{E}(\Delta, p)$) with increment rate of 5.00×10^{-3} and four probability levels defining the likelihood of cooperation ($p = \{0.25, 0.50, 0.75, 1.00\}$). The x-axes correspond to the proportions of cooperators (x_a). The y-axes denote differential equation (8) as an implicit function of time ($E(\Delta, p)/dt$). The model generates three main patterns of population dynamics. The trajectories to fixed points are illustrated through straight lines and periodic orbits, the latter being represented by a series of bell-shaped and of s-shaped curves. They exhibit increasing order close to full competition and cyclic temporary bifurcations toward full cooperation

4.3 Price identity of unstructured population replicator–mutator dynamics

Unstructured population replicator–mutator dynamics (Fig. 3) shows bifurcation patterns that illustrate the introduction of certain mutation toward cooperation.

At $p = 0.25$, be it a low probability of playing against a cooperative model-player, the cyclical shift in perfect order, that is, $\dot{E}(\alpha, 0.25) = 0.00$, is once again validated. The non-monotonic pseudo u-shaped trajectories, with an inflection point near $x_a = 1.00$, is the predominant figurative aspect of the differential equation. The attraction toward a cooperative population configuration, by cause of the introduction of certain mutation of competitive model-players, is found to be in great stability. For example, the maximum value, registered at $x_a = 0.96$, is of $\dot{E}(0.25) = -2,743$. At this value, the endowment attains the value of -30.04 , which implies that the attraction toward full cooperation is obtained at the expense of a triple deficit of wealth. Nonetheless, above this fraction, the arrival at a population setting wholly composed of cooperators switches the sign of the time derivative and it exclusively happens in the positive orthant. Accordingly, the objective of reaching a population-wide cooperation ultimately occurs with at a moderate magnitude of instability. That

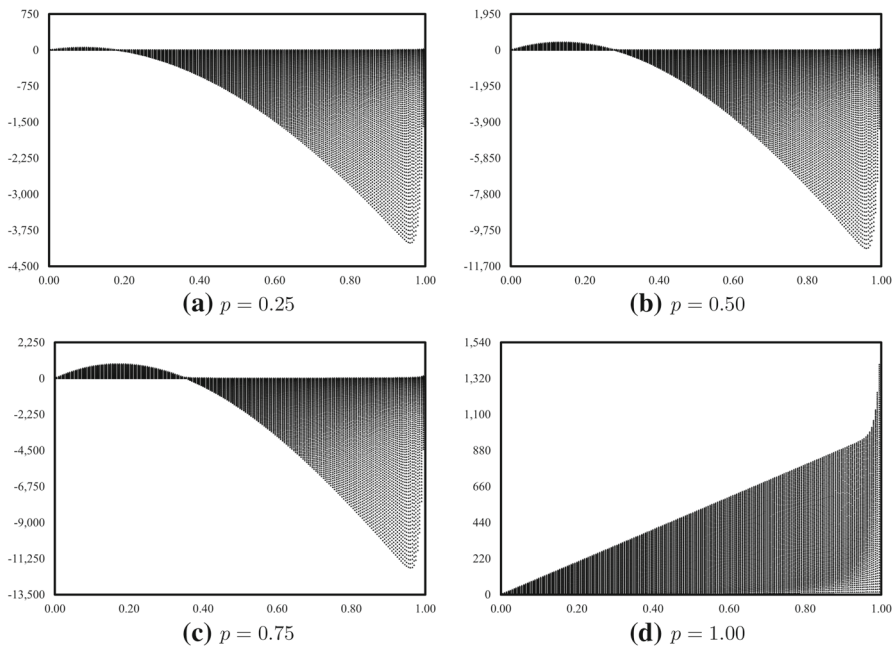


Fig. 4 Bifurcation diagrams of the Price identity applied to structured population replicator–mutator dynamics ($\dot{E}(\alpha, \Delta, p)$) with increment rate of 5.00×10^{-3} and four probability levels defining the likelihood of cooperation ($p = \{0.25, 0.50, 0.75, 1.00\}$). The x -axes correspond to the proportions of cooperators (x_a). The y -axes denote differential equation (10) as an implicit function of time ($E(\alpha, \Delta, p)/dt$). The model generates three main patterns of population dynamics. The trajectories to fixed points are illustrated through straight lines and periodic orbits, the latter being represented by a series of bell-shaped and of s-shaped curves. They exhibit increasing order close to full competition and cyclic temporary bifurcations toward full cooperation

is, $x_a = 1.00$ takes place at $\dot{E}(0.25) = 136$. When the level of cooperation is low ($x_a \rightarrow 0.00$), the likelihood of cooperation is low as well ($p \rightarrow 0.00$), which confirms the result obtained in the N -player game.

Such as previously mentioned, the completion of full cooperation is asymptotic with respect to the time derivative. This is verified by the fact that $\dot{E}(0.25) = 226$ at $x_a = 1.00$. The likelihood of cooperation turning into certainty ($p \rightarrow 1.00$), accompanied by the certainty of mutation ($q_{ji} = 1.00$), impacts the behavior of model-players toward a cooperative behavior. However, the convex-shaped triangular areas, located in the positive orthants, increase in sizes to the detriment of stability toward full cooperation. The most striking illustration comes from the excessive level of $\dot{E}(1.00)$ at $x_a = 1.00$, which is equal to 97,034.

4.4 Price identity of structured population replicator–mutator dynamics

Figure 4 provides an overview of replicator–mutator dynamics within a structured population structure.

The pattern of $\dot{E}(\alpha, \Delta, 0.25) = 0.00$ is again proved true. Unlike the previous scenarios, the main difference in the bifurcation diagrams comes from the imperfect reflection of those observed with structured population replicator dynamics. Indeed, deformed reverse u-shaped and pseudo triangular regions can be noticed. In detail, two intersections with the straight line of perfect cyclical shifts between competition and cooperation should be emphasized. Likewise, the regions in which the attraction toward full competition take place rise up at $p = \{0.25, 0.50, 0.75\}$. In this instance, all basins of attraction toward competition happen to be located in positive orthants, with increasing instability ($\dot{E}(\alpha, \Delta, p) \gg 0.00$) proportional to increasing likelihood of encountering a cooperative model-player ($p \rightarrow 1.00$). When $p = 0.25$ and $x_a = 0.10$, $\dot{E}(\alpha, \Delta, 0.25) = 55$. At these coordinates, the shift to full competition requires a level of wealth of 4.69, such that $w \leq c$. Likewise, $p = 0.25$ and $x_a = 0.96$, $\dot{E}(\alpha, \Delta, 0.25) = -4009$, so that the shift to full cooperation demands an endowment of 151. It amounts to a multiplication of the initial level of wealth by 15.07.

The sizes of basins of attraction toward the extinction of cooperators are significantly disproportioned in regards to the triangular-shaped regions unveiling trajectories toward full cooperation. As an illustrative example apropos of $p = 0.75$, the peak value of $\dot{E}(\alpha, \Delta, 0.75) = 910$ at $x_a = 0.17$ is, in absolute terms, much lower than the extreme value of $\dot{E}(\alpha, \Delta, 0.75) = -11,827$ at $x_a = 0.96$. The final case, where $p = 1.00$, unveils a pattern uniquely present in the positive orthant. The asymptotic convergence toward full density of cooperators, identified at $x_a = 0.96$, unfolds a high swing in instability in that $\dot{E}(\alpha, \Delta, 1.00) = 962$.

4.5 Unified evolutionary dynamics through the Price identity

An additional analysis of the unified evolutionary dynamics, with respect to the Price identity, is now undertaken. As can be noted in Fig. 5, the coupling of unstructured population replicator dynamics with structured population replicator dynamics gives rise to the newly termed joint selection of cooperators and competitors.

At $x_a^* = 0.50$, a proportional relationship between the levels of rewarding ($\delta \in [0.00, 1.00]$) and the joint selection criterion ($M(\delta) = \frac{\pi_a(1.00-x_a)}{\text{cov}(\pi_a, p) + \pi \bar{p}} (\text{cov}(\pi_a + \pi_b, p) + 2\pi \bar{p})$) arises therefrom. The illustration shows that, along the axis of probability levels, the relationship happens to be invariable. In this respect, the selection is unaffected by the likelihood of meeting a cooperative model-player. This is because $2\bar{p}$ outweighs p , within the expression of $M(\delta)$, when $x_a \geq p > 0$. In detail, should the rewarding level be of zero, the selection of cooperators and competitors equals the cost of cooperating ($M(0.00) = 5.00 \equiv c$). When the rewarding meets the value of $\delta = 1.00$, the aggregated selection amounts to the level of wealth ($M(1.00) = 10.00 \equiv w$).

Based on the properties set out above, the Price theorem of selection enables to reach a specific form of collective coordination. The latter states that model-players, regardless of the game structure in which they evolve, can theoretically achieve, under strategic uncertainty and mutation processes, a population-wide cooperation sufficient enough to avoid the tragedy of the commons. Such a configuration is applicable on condition that half of the population, which ought to be rewarded at a level equivalent to the joint selection of players issued from the unifying Price identity, has a cooperative

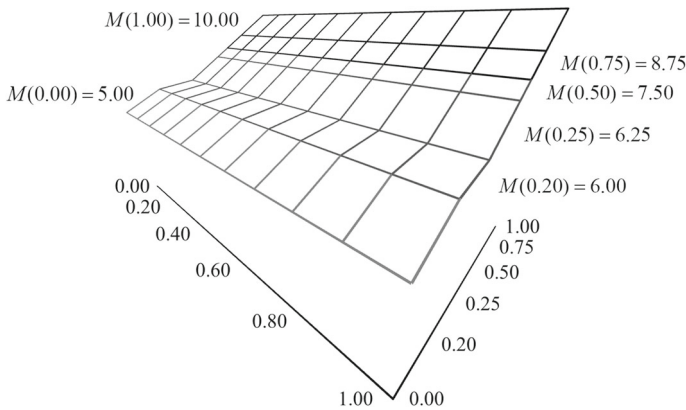


Fig. 5 The z -axis denotes the applicates of $M(\delta) = \frac{\pi_a(1.00-x_a)}{\text{cov}(\pi_a, p) + \pi \bar{p}} (\text{cov}(\pi_a + \pi_b, p) + 2\pi \bar{p})$ that unify $\dot{x}_a$ with $\dot{x}_a(\alpha)$ at $x_a^* = 0.50$. The left-sided abscissa (x -axis) represent the likelihood of cooperation $p \in [0.00, 1.00]$. The right-sided ordinates (y -axis) map various scenarios of rewarding $\delta = \{0.00, 0.20, 0.25, 0.50, 0.75, 1.00\}$

behavior. De facto, our fallouts enrich the literature on collective benefits of group memberships, where the fraction of cooperators ought to exceed a threshold value (Perc 2016; Perc et al. 2017).

5 Conclusion

Such as pointed out by Page and Nowak (2002), the equivalence between the replicator–mutator dynamics and the Price equation is at the center of unified evolutionary dynamics. The modeling framework used in the present paper shows a somewhat different equivalence, in that an aggregated version of the Price formula permits connecting both replicator dynamics and replicator–mutator dynamics through the consideration of the population structure. To the best of our knowledge, this result has not been reported in the existing literature. In addition, it extends the findings in Dragicevic (2019a) on the interlocking between evolutionary dynamics. This notwithstanding, the limits of the approach need to be underlined, for the equivalence comes from a peculiar form of decay factor, be it the portion of homogeneous individuals in a group employing the opposite strategy than that taken heed of. Different forms of structuring should certainly lead to different conclusions. As a consequence, the mainstreaming potential of the outcomes has to be tempered.

With regards to the outputs on cooperation in uncertainty and subject to mutation processes, the key takeaways include greater attraction toward competition and shifts toward full cooperation on a cyclical basis. The emergence of cyclic dominance could be explained as an unavoidable phenomenon of discontinuous phase transitions (Perc 2016). It gives to the population-wide cooperation a transient status. Cyclical-type interactions have already been put forward in public goods games with correlated reward and punishment (Szolnoki and Perc 2013), in ultimatum games (Szolnoki et al. 2012) and in social dilemma games with coevolution (Perc and Szolnoki 2010).

High mutation rates are mostly observed in cyclical frequency-dependent dynamics as well as in fluctuating environments (Allen and Rosenbloom 2012). Taking account of decay factors, given that the spread of a strategy depends on the frequency relative to other strategies borrowed by the population, as well as of a fluctuating environment, the presumption of which can be deduced from the probability change over time (Hilbert 2017), are thus adequate. To quote another result from Hilbert (2017), cooperation has been found to outperform selection when complementary variety between different population types is observed. When put into perspective with our setting, the finding implies that the presence of cooperators and competitors in the Price identity can be interpreted as a complementary selection of population types. If so, cooperation and selection are not necessarily in contradiction.

The enrichment of the debate comes from the perfect equality between differential equations. In particular, all things considered at some point of time, no less than half or a mere half of the population ought to demonstrate a cooperative behavior. This property suggests that the temporary attainment of full cooperation, that is, the absolute avoidance of the tragedy of the commons, could probably not be maintained over a long period. In that matter, the presence of competitors susceptible to mutation is probably a necessary but not a sufficient condition for the cooperation to be meaningful at any point along the evolutionary course. Sufficiency may be achieved through a structured population setting, which instills doubts raised by Hilbe (2011) on the outcomes resulting from well-mixed populations. What the equality also argues is that, in the lack of certainty on whether the population is unstructured or structured, just as much on the type of evolutionary dynamics governing the agents' behaviors, ending up with an unstable half of the population in cooperation is enough to avoid the tragedy related to the overharvesting of common-pool resources.

In summary, in light of the fact that the Guiana Shield represents 26% of Amazonia, the territory of which was (Laurence et al. 2004) and continues to be (Kalamandeen et al. 2018) under threats of deforestation and of fragmentation of the forest cover, the modeling outputs argue for the setting up of additional cooperative projects. Despite the urgent need for a greater regard toward other parts of the Amazon at risk, reaching a total level of cooperation up to 50% of the stakeholders legally assigned responsibility, or willing to hold the responsibility, for their conservation might be sufficient to avoid irreversible damages. In reality, in spite of the existence of local and international organizations dedicated to forest conservation, the driving forces of change that affect forests and their users and beneficiaries interact in complex ways and on various levels. Forasmuch as local protection initiatives lead to cumulative land-use decisions, the latter contribute to both creating and solving global problems (IUFRO 2010). Those implicitly reveal the presence of spatially structured local communities and the absence of a cross-country structured society involved in the safeguard of tropical forests. Likewise, the omnipresence of defectors provoking the forest disappearance, which fortunately or unfortunately encourage cooperation between agents keen to preserve it, implies the certainty to fail in obtaining full cooperation as a steady state. This spurs the countries to look for a centralized transitory leadership that would enable to reach the imperative partnership at the local and global scales.

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Appendix

Proof of Proposition 3 The stability conditions are reviewed through the fixed points. Solving $\dot{E}(p) = 0.00$ yields $x_a = 0.00$ and $x_a = 1.00$. We look at the derivative of $\dot{E}(p)$ labeled $F'(x_a)$. We have

$$\begin{aligned} F'(x_a) = & 2px_a(1.00 - x_a) \left[(N - 1)px_a^{N-1}(w(1.00 + \delta) - c) \right. \\ & \left. + (N - 1)(1.00 - p)wx_a^{N-1} \right] \\ & + p(1.00 - x_a)x_a^2 \left[p(w(1.00 + \delta) - c)x_a^{N-2} + (1.00 - p)(w - c) \right] \\ & + p(1.00 - x_a)x_a^2 \left[-(1.00 - p)w(1.00 - x_a^{N-2}) - pw \right] \\ & - px_a^2 \left[p(w(1.00 + \delta) - c)x_a^{N-1} + (1.00 - p)(w - c) \right] \\ & - px_a^2 \left[-(1.00 - p)w(1.00 - x_a^{N-1}) - pw \right] + \Sigma_i \dot{p}, \end{aligned}$$

where $\dot{p} = n\bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$ for $n \in i = 1, \dots, N$. At $x_a = 0.00$, the Jacobian determinant is $F'(0.00) = J(x_a = 0.00) = 0.00$, which matches with an equilibrium that undergoes cyclical behavior. At $x_a = 1.00$, $F'(1) = J(x_a = 1.00) \geq 0.00$ when $\Sigma_i \dot{p} \geq -p[p(w(1.00 + \delta) - c) + (1.00 - p)(w - c) - pw]$, so the equilibrium can be either stable, unstable, or cyclical. We thus observe cases of cyclical shifts to either extinction or full density of cooperators, with a greater attraction toward competition, for the (in)stability of $F'(0.00)$ is independent of p .

By neglecting higher-order terms raised to power N , for $x_a^N = 0.00$ when $N \rightarrow \infty$, the differential function can be reduced. The resulting formula permits revealing the existence of an interior equilibrium population size (Tenreiro Machado 2018). We observe that

$$\dot{E}(p) = x_a(1.00 - x_a) [(1.00 - p)(w - c) - pw - (1.00 - p)w] \Sigma_i px_a + \Sigma_i x_a \dot{p},$$

where $n \in i = 1, \dots, N$ and $\dot{p} = n\bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$. In such a case, $\Sigma_i x_a = 0.50 \pm \frac{\sqrt{pc(p-1) - p^2w + 4\dot{p}}}{2\sqrt{p}\sqrt{c(p-1) - pw}}$. In spite of that, its existence is compromised unless $c \gg w$, which is little plausible in the real-world context of common-good management. We shall thus inspect the behavior of evolutionary dynamics as if the preminent goal was to obtain a maximum number of cooperators. Consequently, we have $\lim_{x_a \rightarrow 1.00} \dot{E}(p) = \Sigma_i x_a \dot{p} \Leftrightarrow \Sigma_i x_a = N\dot{p}$. Rewriting $\Sigma_i x_a = N\dot{p} \Leftrightarrow x_a = \frac{1}{N}\dot{p}$, for $i = 1, \dots, N$, entails the existence of a barycenter in the probability space (Elqayam and Over 2016). Its evolution decides on the evolution of the cooperative population. The applicability of population dynamics based on model-players can

therefore be justified. All the same, were $\dot{p} \simeq 0.00$, one obtains $x_a \simeq 0.00$. The interior equilibrium should then be near the quasi-absence of cooperators.

Further inquiry of the replicator equation leads us to the study of the inflection points. The latter can be obtained by fixing $F''(x)$ equal to zero, that is, by computing the second derivative. Indeed, the its existence implies the potentiality of presence of inflection points, for second derivatives measure both the curvature of changes in the population growth and in its parameters (Shyu and Caswell 2014). Their placement relative to model-players' levels of wealth determine their risk attitude withal (Gillen and Markowitz 2010). The differentiation yields

$$\begin{aligned}
 F''(x_a) &= 4px_a(1.00 - x_a) \left[p(w(1.00 + \delta) - c)(N - 1)x_a^{N-2} \right] \\
 &= 4px_a(1.00 - x_a) \left[-(1.00 - p)(w - c)(N - 1)x_a^{N-2} \right] \\
 &\quad - 4px_a \left[p(w(1.00 + \delta) - c)x_a^{N-1} + (1.00 - p)(w - c)(1.00 - x_a^{N-1}) \right] \\
 &\quad + 2p(1.00 - x_a) \left[p(w(1.00 + \delta) - c)x_a^{N-1} \right. \\
 &\quad \left. + (1.00 - p)(w - c)(1.00 - x_a^{N-1}) \right] \\
 &\quad + px_a^2(1.00 - x_a) \left[p(w(1.00 + \delta) - c)(N - 2)(N - 1)x_a^{N-3} \right] \\
 &\quad + px_a^2(1.00 - x_a) \left[-(1.00 - p)(w - c)(N - 2)(N - 1)x_a^{N-3} \right] \\
 &\quad - 2px_a^2 \left[p(w(1.00 + \delta) - c)(N - 1)x_a^{N-2} \right. \\
 &\quad \left. - (1.00 - p)(w - c)(N - 1)x_a^{N-2} \right] = 0.
 \end{aligned}$$

Provided the complexity of the equation, a straight conclusion on a level of density, where the curvature switching crops up, cannot be made. The coordinates of the inflection points shall thus be numerically approximated in the sections dedicated to simulation examples. Notwithstanding the inability to obtain such an analytical result, we can scrutinize, through the study of the limit values of two function parameters, the behavior of the second derivative. In that manner, we observe that

$$\left[\begin{array}{l}
 \lim_{x_a \rightarrow 0} F''(x_a) = 0.00, \\
 \lim_{p \rightarrow 0} F''(x_a) = 0.00, \\
 \lim_{x_a \rightarrow 1} F''(x_a) = 4p^2(w(1.00 + \delta) - c) + 2(p(w(1.00 + \delta) - c) \\
 \quad - (1.00 - p)(w - c))(N - 1)p, \\
 \lim_{p \rightarrow 1} F''(x_a) = 4(1.00 - x_a)(w(1.00 + \delta) - c)(N - 1)x_a^{N-1} \\
 \quad + 4(w(1.00 + \delta) - c)x_a^N + \frac{(w(1.00 + \delta) - c)x_a^N}{x_a} \\
 \quad - (w(1.00 + \delta) - c)(N - 2)(N - 1)(x_a - 1)x_a^{N-1} \\
 \quad + 2(w(1.00 + \delta) - c)(N - 1)x_a^N.
 \end{array} \right]$$

When $x_a \rightarrow 1.00$, $F''(x_a) = 4p^2(w(1.00 + \delta) - c) + 2(p(w(1.00 + \delta) - c) - (1.00 - p)(w - c))(N - 1)p = 0.00 \Leftrightarrow p \rightarrow 1.00$. Provided the uncertainty of

having $p = 1.00$, the density of cooperators is confirmed to be susceptible to cyclical shifts. When $p \rightarrow 1.00$, $F''(x_a) = 0.00 \Leftrightarrow x_a = 0.00$ for $N > 1$. Otherwise, $x_a > 0.00$, for $w = \frac{c}{1+\delta}$, such that the level of wealth equals the discounted value of the expense of cooperation. If model-players accept profits as well as losses, they are regarded as neutral against risk (Ramamohan Rao 2016). That is to say, in the presence of risk-neutral model-players, a population-wide cooperation is contingent on their diminished levels of wealth. $\square$

Proof of Proposition 4 This time, root $x_a = 0.00$, as a fixed point of replicator dynamics applied to a structured population setting, goes along with $pw = -(1.00 - p)w \Leftrightarrow w = 0.00$. The derivative is as follows

$$\begin{aligned} G'(x_a) = & (N-1)(p(w(1.00 + \delta) - pw))x_a^{N-2} \\ & + 2x_a((1.00 - p)(w - c) - pw - (1.00 - p)w) \\ & + 2(pw + (1.00 - p)w) + (1.00 - p)w(N-1)x_a^{N-2} \\ & + \Sigma_i \dot{p}, \end{aligned}$$

where $\dot{p} = n\bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$ for $n \in i = 1, \dots, N$. At $x_a = 0.00$, the Jacobian determinant is $G'(0.00) = J(x_a = 0.00) = 2(pw + (1.00 - p)w) + \Sigma_i \dot{p} \gtrless 0.00$, which means that the equilibrium can be stable, unstable or cyclical. In addition, should $\Sigma_i \dot{p} = 0.00$, with zero frequency-dependent payoff variation—included within the probability measure, the nature of the equilibrium dynamics depends on p and on w .

Through the study of the limit values of two main function parameters, the behavior of the second derivative can be studied. The differentiation yields

$$\begin{aligned} G''(x_a) = & (p(w(1.00 + \delta) - pw)(N-2)(N-1)x_a^{N-3} \\ & + 2((1.00 - p)(w - c) - (1.00 - p)w - pw) \\ & + (N-2)(N-1)(1.00 - p)wx_a^{N-3}. \end{aligned}$$

Further investigation of the replicator equation, under a structured population setting, is conducted through that of the inflection points.

$$\left[\begin{array}{l} \lim_{x_a \rightarrow 0} G''(x_a) = 2((1.00 - p)(w - c) - (1.00 - p)w - pw), \\ \lim_{p \rightarrow 0} G''(x_a) = 2((w - c) - w - pw) + (N-2)(N-1)x_a^{N-3}, \\ \lim_{x_a \rightarrow 1} G''(x_a) = (p(w(1.00 + \delta) - pw)(N-2)(N-1) \\ \quad + 2((1.00 - p)(w - c) - (1.00 - p)w - pw) \\ \quad + (N-2)(N-1)(1.00 - p)w), \\ \lim_{p \rightarrow 1} G''(x_a) = (w(1.00 + \delta) - w)(N-2)(N-1)x_a^{N-3}. \end{array} \right]$$

When $x_a \rightarrow 0.00$, $G''(x_a) = 2((1.00 - p)(w - c) - (1.00 - p)w - pw) = 0.00 \Leftrightarrow p = \frac{c}{c-w}$. Owing to the fact that $c \leq w$, the theoretical result is hardly applicable in reality. In the same way, $x_a \rightarrow 1.00$ yields $p = 0.00$. In contrast, $p \rightarrow 0.00$

leads to having $x_a = 2^{-\frac{1}{3-N}} \sqrt[3-N]{\frac{N^2-3N+2}{c+pw}} \Leftrightarrow x_a \rightarrow 1.00$. When $p \rightarrow 1.00$, we have $x_a = 0.00$ for $N > 0$ and $x_a > 0.00$ for $w = 0.00$. Would the model-players' levels of wealth turn out to be of zero, a coalition between cooperative strategists could possibly be gained. On account of $w(1.00 + \delta) - c = 0.00$, a loss leads to either having $w = \frac{c}{1+\delta}$, such that the level of wealth equals the discounted value of the expense of cooperation, or to having $w = 0.00$, such that model-players are left with null endowments. $\square$

Proof of Proposition 5 The root of the differential equation is equal to $x_a = 0.00$. The differentiation of the equation gives the following result

$$\begin{aligned} H'(x_a) = & -Npx_a \left((p(w(1.00 + \delta) - c) x_a^{N-1} + (1.00 - p)(w - c)) \right) \\ & + Np(1.00 - x_a) \left((p(w(1.00 + \delta) - c) x_a^{N-1} + (1.00 - p)(w - c)) \right) \\ & + (p(w(1.00 + \delta) - c) (N - 1)Np(1.00 - x_a) x_a^{N-1} \\ & - (pw - (1.00 - p)w) (N - 1)px_a^{N-2} (x_b N - N(1.00 - x_a)x_a) \\ & + (pw - (1.00 - p)w) p (Nx_a - N(1.00 - x_a)) (1.00 - x_a^{N-1}) + \sum_i \dot{p}, \end{aligned}$$

where $\dot{p} = n\bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$ for $n \in i = 1, \dots, N$. At $x_a = 0.00$, the Jacobian determinant is $H'(0.00) = J(x_a = 0.00) = N(-p(w + c) - p^2(w - c) - w - \dot{p}) < 0.00$. That is, the only root is a steady state. In greater detail, $N(-p(w + c) - p^2(w - c) - w - \dot{p}) < 0.00 \Leftrightarrow w > \frac{c(p^2 - p) - \dot{p}}{p(p+1)+1}$ for $N > 0$ and $w \geq c$. In presence of mutation, the results show that the replicator-mutator dynamics corresponds to a stable equilibrium in which competition unconditionally attracts the model-player. Despite the fact that the derivative does depend on the probability to engage into a cooperative behavior, its presence only attenuates the attraction toward the competitive equilibrium.

The second derivative of the differential equation yields

$$\begin{aligned} H''(x_a) = & -(N - 1)Np^2x_a^{N-1}((1 + \delta)w - c) \\ & - Np(px_a^{N-1}((1 + \delta)w - c) + (1.00 - p)(w - c)) \\ & + (N - 1)Np^2(1.00 - x_a)x_a^{N-2}((1 + \delta)w - c) \\ & - Np(px_a^{N-1}((1 + \delta)w - c) + (1.00 - p)(w - c)) \\ & + (N - 1)^2Np^2(1.00 - x_a)x_a^{N-2}((1 + \delta)w - c) \\ & - (N - 1)Np^2x_a^{N-1}((1 + \delta)w - c) \\ & - 2(N - 1)(p - 0.50)px_a^{N-3}(Nx_a^2 + (1 - N)x_a + (N - 2)Nx_b) \\ & + (N + 1)p(1.00 - x_a^{N-1})(pw - (1.00 - p)w) \\ & - (N - 1)px_a^{N-2}((N + 1)x_a - 1)(pw - (1.00 - p)w). \end{aligned}$$

We now initiate the study of the inflection points.

$$\left[\begin{array}{l} \lim_{x_a \rightarrow 0.00} H''(x_a) = 2Np((1.00 - p)(w - c)) + (N + 1)p(pw - (1.00 - p)w), \\ \lim_{p \rightarrow 0.00} H''(x_a) = 0.00, \\ \lim_{x_a \rightarrow 1.00} H''(x_a) = -Np(p((1 + \delta)w - c) + (1.00 - p)(w - c)) \\ \quad - Np(p((1 + \delta)w - c) + (1.00 - p)(w - c)) \\ \quad + (N - 1)^2 Np^2((1 + \delta)w - c) \\ \quad - 2(N - 1)(p - 0.5)pw(N - 2)Nx_b \\ \quad - (N - 1)pN(pw - (1.00 - p)w), \\ \lim_{p \rightarrow 1.00} H''(x_a) = -(N - 1)Nx_a^{N-1}(w(1.00 + \delta) - c) - N(x_a^{N-1}(w(1.00 + \delta) - c)) \\ \quad + (N - 1)N(1.00 - x_a)x_a^{N-2}(w(1.00 + \delta) - c) - N(x_a^{N-1}(w(1.00 + \delta) - c)) \\ \quad + (N - 1)^2 N(1.00 - x_a)x_a^{N-2}(w(1.00 + \delta) - c) - (N - 1)Nx_a^{N-1}(w(1.00 + \delta) - c) \\ \quad - (N - 1)wx_a^{N-3}(Nx_a^2 + (1 - N)x_a + (N - 2)Nx_b) \\ \quad + (N + 1)(1.00 - x_a^{N-1})w - (N - 1)x_a^{N-2}((N + 1)x_a - 1)w. \end{array} \right]$$

When $x_a \rightarrow 0.00$, $H''(x_a) = 2Np((1.00 - p)(w - c)) + (N + 1)p(pw - (1.00 - p)w) = 0.00 \Leftrightarrow p = \frac{2cN - w(N - 1)}{2(cN + w)} \rightarrow 0.00$ for $c \leq w$. Put differently, in case of small chance of encountering a cooperator, the magnitude of the attraction basin toward competition proves to be preponderant. Hence, the result is compatible with the finding that mutation mostly arises when $x_a \in [0.00, 0.50]$ (Barton and Polechová 2005; Garcia and Traulsen 2012). When $x_a \rightarrow 1.00$, we have

$$p = - \frac{-2(c + w) - Nw + N^2w + 2Nwx_b - 3N^2wx_b + N^3wx_b}{2(-AN - 2AN^2 + AN^3 + 2N^3 + 2Nw - 2N^2w - 4Nwx_b + 6N^2wx_b - 2N^3wx_b)} \\ - \frac{-2(c + w) - Nw + N^2w + 2Nwx_b - 3N^2wx_b + N^3wx_b}{4(-AN - 2AN^2 + AN^3 + 2N^3 + 2Nw - 2N^2w - 4Nwx_b + 6N^2wx_b - 2N^3wx_b)} \\ - \left[\frac{2(w - c)}{-AN - 2AN^2 + AN^3 + 2Nw - 2N^2w - 4Nwx_b + 6N^2wx_b - 2N^3wx_b} \right]^{0.50},$$

with $A = w(1.00 + \delta) - c$, which gives rise to $p = (0.50, 1.00]$. Whenever the population pattern heads in the direction of full density of cooperators, engaging into a cooperative partnership inevitably occurs with significant likelihood. As for $p \rightarrow 1.00$, we obtain $x_a = 0.00$. That is, despite the certainty of facing a cooperative model-player, the cooperative fraction vanishes. This is due to the fact that x_a continues to be pulled back toward a competitive behavior in a cyclical manner. Hence, the full cooperation is inevitably transient. $\square$

Proof of Proposition 6 When $E(\Delta) = 1.00$, the likelihood of meeting a cooperator becomes great ($p \rightarrow 1.00$). We assume that $\Sigma_i x_a \pi_a p \equiv \text{cov}(\pi_a, p) + \bar{\pi} \bar{p}$ and $\Sigma_i (1.00 - x_a) \pi_b p \equiv \text{cov}(\pi_b, p) + \bar{\pi} \bar{p}$. Provided that $x_a \in [0.00, 1.00]$, the previous equation engenders the upcoming constraints. In order to obtain $x_a \geq 0.00$, both expressions of the fraction ought to have the same sign. When $\text{cov}(\pi_a, p) + \bar{\pi} \bar{p} \neq 0.00$, this is true for $w \geq c$. In parallel, $x_a \leq 1.00$ leads to verifying, for $\text{cov}(\pi_a, p) + \bar{\pi} \bar{p} \neq$

0.00, $\pi_a(1.00 - x_a)(\text{cov}(\pi_a + \pi_b, p) + 2\bar{\pi}\bar{p}) \leq \text{cov}(\pi_a, p) + \bar{\pi}\bar{p}$, which holds for $p \rightarrow 1.00$. This gives rise to considering the expected payoff of the cooperative model-player, for $\text{cov}(\pi_a, p) + \bar{\pi}\bar{p} = \Sigma_i x_a \pi_a p - \bar{\pi}\bar{p} + \bar{\pi}\bar{p} = \Sigma_i x_a \pi_a p$. Provided that $x_a \geq 0.00$ is verified by assumption, the inequality discloses that, in case of certain mutation, the fraction of competitors, weighted by the payoff of a cooperative model-player and inversely proportioned to the expected payoff from cooperation, should be less than or equal to the selection exerted on both cooperators and competitors increased by the double average population payoff. For the purpose of giving a formalized meaning to the right-hand expression in (9), that is, $\frac{\pi_a(1.00-x_a)}{\text{cov}(\pi_a, p) + \bar{\pi}\bar{p}}(\text{cov}(\pi_a + \pi_b, p) + 2\bar{\pi}\bar{p})$, we have decided to name it joint selection of cooperators and competitors. $\square$

Proof of Proposition 7 The root of the differential equation is equal to $x_a = 0.00$. The differentiation of the equation gives the following result

$$\begin{aligned} I'(x_a) = & -Np \left((p((N-1)(w(1.00 + \delta) - c)) p x_a^N + 2(1.00 - p)(w - c)x_a) \right. \\ & - 2N(1.00 - p)p^2 w(1.00 - x_a)(1.00 - x_a^{N-1}) \\ & \left. + (N-1)p(1.00 - p)w x_a^{N-2}(N x_b - Np(1.00 - x_a)^2) + \Sigma_i \dot{p}, \right. \end{aligned}$$

where $\dot{p} = n\bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$ for $n \in i = 1, \dots, N$. At $x_a = 0.00$, the Jacobian determinant is $H'(0.00) = J(x_a = 0.00) = -2N(1.00 - p)p^2 + \Sigma_i \dot{p} \leq 0.00$. In presence of replicator-mutator dynamics with update decay, the outcomes reveal the possibility of a stable equilibrium subject to cyclical shifts. Even if the derivative depends on the levels of p , its stochasticity only attenuates the attraction toward the competitive equilibrium of the population of model-players.

Ignoring the higher-order terms raised to power N permits studying the interior equilibrium population density. The function at stake becomes

$$\begin{aligned} I(x_a) = & -\Sigma_i x_a p(1.00 - p)(w - c) \\ & + (pw - (1.00 - p)w) p \left(\Sigma_j x_b - \Sigma_i (1.00 - x_a^2) p \right) + \Sigma_i x_a \dot{p}, \end{aligned}$$

where $i, j = 1, \dots, N$ and $\dot{p} = n\bar{p} \left(N + 5.00 \times 10^{-3} \frac{n}{N} \right) \frac{\pi_a(n) - \pi_b(n)}{\pi_a(n-1) - \pi_b(n-1)}$ for $n \in i = 1, \dots, N$. Setting the differential equation equal to zero yields

$$\Sigma_i x_a = - \frac{p(2p - 1)w(\Sigma_i p - \Sigma_j x_b)}{c(p - 1)p + (2p^2 - 2p + 1)pw - \dot{p}}.$$

The second derivative of the differential equation yields

$$\begin{aligned} I''(x_a) = & -Np((N-1)Np^2 x_a^{N-1}((1 + \delta)w - c) + 2(1.00 - p)(w - c)) \\ & + 2(N-1)N(1.00 - p)p^2 w(1.00 - x_a)x_a^{N-2} \\ & + 2N(1.00 - p)p^2 w(1.00 - x_a^{N-1}) \\ & + 2(N-1)N(1.00 - p)p^2 w(1.00 - x_a)x_a^{N-2} \\ & + (N-2)(N-1)(1.00 - p)pw x_a^{N-3}(N x_b - Np(1.00 - x_a)^2). \end{aligned}$$

The differentiation of the equation provides us with the opportunity to study the existence of the inflection points.

$$\left[\begin{array}{l} \lim_{x_a \rightarrow 0.00} I''(x_a) = -2Np((1.00 - p)(w - c)) + 2N(1.00 - p)p^2w, \\ \lim_{p \rightarrow 0.00} I''(x_a) = 0.00, \\ \lim_{x_a \rightarrow 1.00} I''(x_a) = -Np((N - 1)Np^2((1 + \delta) - c)) + 2(1.00 - p)(w - c) \\ \quad + (N - 2)(N - 1)(1.00 - p)pwNx_b, \\ \lim_{p \rightarrow 1.00} I''(x_a) = -N((N - 1)Nx_a^{N-1}((1 + \delta) - c)) + 2(1.00 - p)(w - c). \end{array} \right]$$

When $x_a \rightarrow 0.00$, $I''(x_a) = -2Np((1.00 - p)(w - c)) + 2N(1.00 - p)p^2w = 0.00 \Leftrightarrow p = \{0.00, \frac{w-c}{w}, 1.00\}$ for $c \leq w$. Ergo, whatever the probability of meeting a cooperator, the attraction of the competitive behavior at the population scale appears unavoidable. When $x_a \rightarrow 1.00$, we have $p = 1.00$ for $c = 1.00 + \delta$. In other terms, full cooperation can be achieved were its cost be equal to the rewarding multiplier. When it comes to $p \rightarrow 1.00$, the inflection point is supposed to take place at $x_a = 0.00$, for $N > 1$, and at some $x_a > 0.00$, for $c = 1.00 + \delta$. $\square$

Proof of Proposition 8 Equivalent to that of Proposition 6. $\square$

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